

The Heavens Scaled Down

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Abstract

The goal is to build a scaled down Solar System with the boundaries from the Earth and its moon, Luna, to the Sun. Obviously, gravity will be a huge part of this system; the intention is to use Lagrangians instead of Newtonian mechanics, especially because the book has some references to general relativity. The codebase will be constructed from Ada for the system, Python for scripting, and a Bash environment. Documentation will be done with $\text{T}_{\text{E}}\text{X}$. This will be a cross collaboration between the tools of Claude AI and Replit. To start, Chapter 2.2 and Appendix C of *Statistical Orbit Determination (2004)* by Byron Tapley, Bob Schutz, & George H. Born will be noted. **Then, ...**

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1 Introduction

2 The Fundamental Mechanics

This section explores the fundamental math and physics required for the proposed system. Particularly, §2.1 contains notes on Chapter 2.2 of [Tapley et al. \(2004\)](#), and §2.2 is a summary of its Appendix C [1].

2.1 General Properties of the Two-Body Problem

Although a point mass Earth is a serious stretch of the imagination, a spherical body with constant, uniform mass distribution can be shown to be gravitationally equivalent to a point mass. Thus, as a more realistic approximation of the physical characteristics of a planet, a spherical body will be used.

Consider the motion of two spheres of mass M_1 and M_2 , as illustrated in Figure 2.1. No restrictions are placed on the magnitude of these masses; so, for illustrative purposes, assume M_1 represents the Earth and M_2 represents a satellite, such as Luna. The coordinate system is inertial and nonrotational. The equations of motion derived from Newton's Laws of Motion and Law of Gravitation are:

$$M_1 \ddot{\vec{R}}_1 = \frac{GM_1 M_2}{r^3} \vec{r} \quad (2.1a)$$

$$M_2 \ddot{\vec{R}}_2 = -\frac{GM_1 M_2}{r^3} \vec{r}. \quad (2.1b)$$

It follows that,

$$\ddot{\vec{r}} = -\frac{\mu}{r^3} \vec{r}, \quad (2.2)$$

where $\mu = G(M_1 + M_2)$ is the (*standard*) *gravitational parameter*. Even if M_2 were to be a large satellite such as Luna, then $M_{\text{Earth}} \sim M_{\text{Luna}} \times 10^{1.9}$; hence, $\mu \simeq GM_1$. Moreover, the general characteristics of the motion follow from manipulation of Equations (2.1).

The motion of two-body system is defined by the position of the center of mass of the two spheres, $\vec{R}_{\text{cm}}$;

$$\vec{R}_{\text{cm}} = \frac{M_1 \vec{R}_1 + M_2 \vec{R}_2}{M_1 + M_2}. \quad (2.3)$$

Adding Equation (2.1a) and Equation (2.1b) gives:

$$\ddot{\vec{R}}_{\text{cm}} = 0 \quad (2.4a)$$

$$\vec{R}_{\text{cm}} = \vec{C}_1(t - t_0) + \vec{C}_2, \quad (2.4b)$$

where $\vec{C}_1$ and $\vec{C}_2$ are constants of motion. Equation (2.4b) states that the center of mass of the two spheres moves in a straight line with constant velocity. Note that the straight line motion of $\vec{R}_{\text{cm}}$ is defined in an inertial coordinate system.

As for the angular momentum, it is obvious from Equation (2.2) that $\vec{r} \times \ddot{\vec{r}} = 0$; so,

$$\frac{d}{dt}(\vec{r} \times \dot{\vec{r}}) = 0. \quad (2.5)$$

Define $\vec{h} \equiv \vec{r} \times \dot{\vec{r}}$, the *angular momentum per unit mass* (or *specific angular momentum*); $\vec{h}$ is called the angular momentum per unit mass because it is the angular momentum vector, $L = \vec{r} \times M \dot{\vec{r}}$, divided by M , which normalizes the angular momentum vector to a mass equal to one. Anyway, from Equation (2.5), $\vec{h}$ is a **constant** with respect to time. An important interpretation of this is that the orbital motion is planar, since the $\vec{h}$ vector must be perpendicular to the plane created by the $\vec{r}$ and $\dot{\vec{r}}$ vectors. The plane of motion is referred to as the *orbit plane*. Note that the specific angular momentum is defined for the motion of M_2 with respect to M_1 .

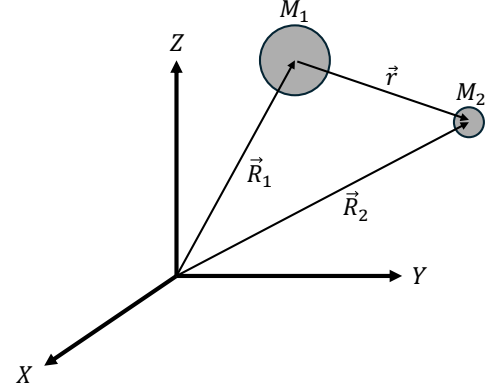


Figure 2.1: M_1 and M_2 are spheres of constant, uniform density. The position vectors of the spheres refer to their respective geometrical center, which also coincide with the center of mass of the sphere.

As for energy, consider the dot product of $\vec{r}$ with Equation (2.2):

$$\dot{\vec{r}} \cdot \ddot{\vec{r}} = -\mu \dot{\vec{r}} \cdot \frac{\vec{r}}{r^3}, \quad (2.6)$$

by the chain rule, the LHS can be simplified to:

$$\dot{\vec{r}} \cdot \ddot{\vec{r}} = \frac{d}{dt} \left(\frac{\dot{\vec{r}} \cdot \dot{\vec{r}}}{2} \right). \quad (2.7)$$

On the other hand, note the following:

$$\frac{d}{dt}(r^{-1}) = \frac{dr^{-1}}{dr} \frac{dr}{dt} = -r^{-2} \frac{dr}{dt}, \quad (2.8)$$

then, recall that $r = \|\vec{r}\| = \sqrt{\vec{r} \cdot \vec{r}}$; so, calculating dr/dt ;

$$\frac{dr}{dt} = \frac{d}{dt} \left(\sqrt{\vec{r} \cdot \vec{r}} \right) \quad (2.9)$$

$$= \frac{1}{2\sqrt{\vec{r} \cdot \vec{r}}} (\dot{\vec{r}} \cdot \vec{r} + \vec{r} \cdot \dot{\vec{r}}) \quad (2.10)$$

$$= \frac{1}{2\sqrt{\vec{r} \cdot \vec{r}}} (2\vec{r} \cdot \dot{\vec{r}}) \quad (2.11)$$

$$= \frac{\vec{r} \cdot \dot{\vec{r}}}{r}. \quad (2.12)$$

Putting (2.12) into Equation (2.8) gives:

$$\frac{d}{dt} \left(\frac{1}{r} \right) = -\frac{\vec{r} \cdot \dot{\vec{r}}}{r^3} \quad (2.13)$$

Now, Equation (2.6) can be written as:

$$\dot{\vec{r}} \cdot \ddot{\vec{r}} = \mu \frac{d}{dt} \left(\frac{1}{r} \right), \quad (2.14)$$

Equating the RHS of Equation (2.14) with the RHS of Equation (2.7) and integrating with respect to t leads to:

$$\xi = \frac{\dot{\vec{r}} \cdot \dot{\vec{r}}}{2} - \frac{\mu}{r}, \quad (2.15)$$

where ξ is mathematically a **constant** of integration, and physically, it is the scalar *energy per unit mass* (also known as *specific energy*). In total, there are ten constants of motion: $\vec{C}_1$, $\vec{C}_2$, $\vec{h}$, and ξ .

2.1.1 Motion in a Plane

2.2 Lagrangian Equations of Motion

References

- [1] Tapley B. D., Schutz B. E., & Born G. H. (2004). Statistical Orbit Determination, *Elsevier Academic Press*. [ISBN-13: [9780126836301](#), ISBN-10: [0-12-683630-2](#)]